INCH-POUND

MIL-M-44342
29 December 1988
SUPERSEDING
EE-M-002142A (Army-GL)
30 April 1986

MILITARY SPECIFICATION

MACARONI AND CHEESE, ARTIFICIALLY COLORED, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers macaroni and cheese, artificially colored, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

| | |
|---|---|
| MIL-L-1497 | - Labeling of Metal Cans for Subsistence Items |
| MIL-L-35078 | - Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for |
| MIL-C-44340(GL) | - Can, Tray Pack |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-900 | - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

United States Standards for Condition of Food Containers

(Copies are available from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)



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U.S. Standards for Grades of Butter

U.S. Standards for Grades of Cheddar Cheese

(Copies are available from the Dairy Standardization Section, Diary Division P.O. Box 96456, Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)



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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Macaroni, elbow. Elbow macaroni shall conform to the FDA Standard of Identity for enriched macaroni products except that semolina flour shall be the only flour used and the diameter of the macaroni shall range from 0.16 to 0.20 inches. The outer curvature length of the macaroni shall range from 0.75 to 1.0 inches and the wall thickness shall range from 0.034 to 0.048 inches. The macaroni shall contain glycerol monostearate, shall have a moisture content not to exceed 13.0 percent, and shall comply with the requirements of MIL-STD-900.

3.2.2 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.



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3.2.3 Cheese, processed, pasteurized, american, white. The pasteurized processed american cheese shall conform to the Food, Drug and Cosmetic Act Standards of Identity for Pasteurized Processed Cheese (21 CFR 133.169) except the cheese shall have a maximum of 40 percent moisture, a maximum of 2.5 percent salt, and the solids shall contain not less than 50 percent fat. The cheese shall possess a pleasing and desirable mild to medium cheese flavor which is characteristic of the variety or varieties of cheese ingredients added. The product may have a slight cooked, acid, or very slight emulsifier flavor, but it shall be free from any undesirable flavors and odors. The pasteurized american cheese shall have a medium firm, smooth and velvety body, free from uncooked cheese particles. It shall be resilient and not tough, brittle, short, weak, or sticky. The cheese shall be white in color and free from mold and burnt particles.

3.2.4 Butter, lightly salted. The lightly salted butter shall be U.S. Grade A or better of the U.S. Standards for Grades of Butter.

3.2.5 Cheese, cheddar, aged. dehydrated, uncolored. Dehydrated cheddar cheese shall be derived from aged cheddar cheese. The dehydrated cheese shall be white in color, uniform in composition and appearance, and free from burnt specks and from lumps that do not fall apart under light finger pressure. The cheese shall possess a characteristic flavor and odor. The dehydrated cheese shall contain not more than 4.0 percent moisture, not more than 6.0 percent salt, and the solids contain not less than 50.0 percent milkfat.

3.2.6 Starch, waxy maize, modified. The starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.7 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.8 Sugar, white, granulated. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.9 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.10 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.



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3.2.11 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The white pepper shall contain not less than 1.0 mL volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.12 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend or powder derived from the endosperm of the seed of Brassica hirta, Moench and Brassica juncea. Mustard flour shall contain not less than 0.4 percent volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.13 Starch, waxy maize, modified (filling and processing aid). Starch shall be a white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hydrated, at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial visosity of the paste shall be retained under conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland, shall demonstrate easy dispersibility and shall comply with MIL-STD-900.

3.2.14 Color, annatto, dry. Annatto color shall be a reddish-orange water soluble powder of the extractive of Annatto bixa Orellana. The powder shall contain not more than 5.0 percent moisture and 100 percent of the powder shall pass through a U.S. Standard No. 10 sieve. The total color (as percent Norbixin) shall be 15 percent.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and further processing.

3.3.1 Macaroni preparation. The macaroni shall be blanched in boiling water until the weight increases by approximately 2.5 times or for sufficient time to assure compliance with finished product requirements. The macaroni shall be immediately cooled to less than 90°F and rinsed to remove excess starch. If the macaroni is not immediately filled into tray pack cans, it shall be cooled to 40°F. The macaroni shall be placed in trays within 2 hours after blanching has been accomplished.



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3.3.2 Sauce preparation. The sauce shall be formulated as follows 1/:

| Ingredient | Percent |
|---|---|
| Water | 69.310 |
| Cheese, processed, pasteurized, american, white | 18.400 |
| Butter, lightly salted | 4.600 |
| Cheese, dehydrated, cheddar | 4.000 |
| Starch, waxy maize modified | 1.500 |
| Salt | 1.000 |
| Sugar, white | .550 |
| Lecithin | .253 |
| Onion powder | .250 |
| Pepper, white | .080 |
| Mustard, flour | .050 |
| Annatto, dry | .007 |
| | 100.00 |

1/ Not more than 2 percent filling and processing aid starch may be used (see 3.2.13 and 6.4).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. Add butter, cheeses, and approximately 2/3 of the water to the processing vessel. Heat with agitation to 180°F.

b. Add salt, sugar, lecithin, onion powder, pepper, and mustard flour to the mixture with constant agitation.

c. Prepare a thin starch slurry using the remainder of the water and add slowly with constant agitation.

d. Heat entire contents to 185°F and hold for 5 minutes.

e. Add annato and blend until color of sauce is uniform.

f. The sauce shall be filled within 4 hours after cooking and held at a temperature of 150° to 180°F during this interval.

g. The volume of the final mixture shall be adjusted to compensate for evaporation loss during heating and holding.



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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with blanched macaroni and sauce such as to conform to the finished product requirements and to the following requirements:

a. A single or multiple stage filling operation may be used.

b. When a multiple stage filling operation is used, the temperature of the sauce shall be 150° to 180°F at time of filling.

c. When a multiple stage filling operation is used, the temperature of the macaroni at time of filling shall be in the temperature range of 28°F to 40°F unless filling occurs immediately after blanching.

d. When a single stage filling operation is used, the temperature of the product shall be 150° to 180°F at time of filling.

e. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirement (see 3.6.k).

f. Each filled and sealed tray pack can shall be in the retort process within 2 hours after filling.

g. When a multiple stage filling operation is performed, thermoprocessing shall be accomplished in a rotary retort system only.

h. When a single stage filling operation is performed, thermoprocessing may be accomplished in a stationary or rotary retort system.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F until a sterilization value ($F_o$) of not less than 6.0 has been achieved. The filled and sealed thermoprocessed tray pack cans shall show no evidence of can swelling when tested for sterility as specified in 4.5.3.4.

3.6 Finished product. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, glass, dirt, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, moldy, rancid, sour, or stale.

c. There shall be no color foreign to the product.

d. The sauce shall not be thin, thick, lumpy, or oily.

e. The sauce shall have a typical processed cheese flavor.



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f. The macaroni shall not be dry, rubbery, mushy, or crumbly.

g. The salt content shall be not less than 0.5 percent nor greater than 1.5 percent.

h. No individual can shall weigh less than 104 ounces.

i. The average net weight shall be not less than 106 ounces.

j. No individual can shall contain less than 60 ounces nor more than 80 ounces of washed drained weight of macaroni.

k. Filled, sealed, and retorted cans must show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

l. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this specification.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).



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4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.



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4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated 40 to 45 minutes in boiling water, opened, and inspected for defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

| Lot size (cans) | Sample size (cans) | Cumulative sample | Acceptance number | Rejection number |
|---|---|---|---|---|
| 0 to 3200 | 8 | -- | 0 | 2 |
| | 8 | 16 | 1 | 2 |
| 3201 to 35000 | 13 | -- | 0 | 3 |
| | 13 | 26 | 3 | 4 |

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product inspection 1/ 2/

| Category | Defect |
|---|---|
| Major | |
| 101 | Washed and drained weight of macaroni in a can is less than 60 ounces or more than 80 ounces. 3/ |
| 102 | Sauce thin, thick, lumpy, or oily. |
| 103 | Sauce does not have a typical processed cheese flavor. |
| 104 | Macaroni texture dry, rubbery, mushy, or crumbly. |
| 105 | Product shows evidence of excessive heating (materially darkened or scorched). |



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1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine washed and drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify the sauce without breaking the macaroni pieces. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the macaroni. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain at least 2 minutes before determining the washed and drained weight by subtracting the sieve tare from the gross weight. The washed and drained weight shall be reported to the nearest 0.1 ounce.

4.5.3.3 Salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by the AMS agent as reserve samples

The three tray packs for laboratory analyses shall be individually tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists except that preparation of the samples shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. Any test failure shall be classified as a major defect. The lot shall be rejected if the salt content of any individual tray pack is less than 0.5 or more than 1.5 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve sample is analyzed, the analysis for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.



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4.5.3.4 Commercial sterility testing. The sample size shall be one filled, sealed, and thermoprocessed tray pack can selected from each retort batch in the lot. The sample cans shall be tested for sterility by incubating the cans at 95° ± 5°F for 10 days. Any evidence of can swelling shall be classified as a critical defect and shall be cause for rejection of the lot.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a critical defect and shall be cause for rejection of the lot:

Evidence of buffing causing scratches or scuffing of exterior can coating.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, i.e., label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D-3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D. Any non-conformance based on observation of can seam teardowns or of record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.



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4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. Buffing of the can causing damage to the exterior can coating is not permitted (see 4.5.4).

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.



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5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three letter code identifying the establishment.
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year. (Example, March 19, 1987 would be coded as 7078.) The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:



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TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

As an alternate labeling method, a preprinted self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
f. Level of packing required (see 5.2).
g. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).



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6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Starch, filling and processing aid. It has been found that Shur-fil 327 starch manufactured by A.E. Staley Mfg. Co., Decatur, IL 62525 meets the requirements of 3.2.1.3 and performs satisfactorily in this product.

6.5 Subject term (key word) listing.

Canned foods
Cheese
Macaroni
Operational rations
Thermostabilized
Tray pack

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SA
Air Force - 50

Review activities:
Army - MD, TS
Navy - MC
DP - SS

Preparing Activity:
Army - GL

(Project 8940-0633)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-44342 | Macaroni and Cheese, Artificially Colored, Thermostabilized, Tray Pack |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION (Mark one) |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER (Specify): _____ |
| b. ADDRESS (Street, City, State, ZIP Code) | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER (Last, First, MI) – Optional | b. WORK TELEPHONE NUMBER (Include Area Code) – Optional |
|---|---|
| c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional | 8. DATE OF SUBMISSION (YYMMDD) |

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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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